Unit 6: Climate Change

Unit 6: Climate Change



At the end of this unit, the learner will be able to:

- Explore the key scientific concepts of climate change.
- Analyze the causes of climate change and how human activities affect the climate.
- Discuss the effects of climate change on living things
- Elaborate the measures to combat climate change



Keyword

Climate change is a long-term change in the earth's climate, especially a change due to an increase in the average atmospheric temperature.

6.1. Climate Change: Causes and effects

6.1.1. Definition of Climate Change

Climate change is the global phenomenon of climate transformation characterized by the changes in the usual climate of the planet (temperature, precipitation, and wind) that are especially caused by human activities. Climate change is a systematic change in the long-term state of the atmosphere over multiple decades or longer.

6.1.2. Causes of climate change

What Causes Climate change?

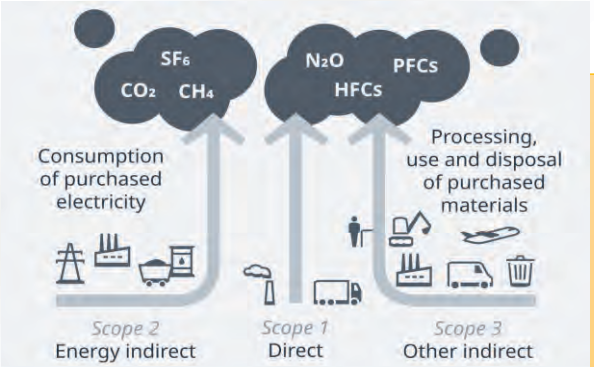


Activity 6.1

Make a small group and discuss about weather, climate, climate variability and climate change. Reflect your answer to the class.

Humans are increasingly influencing the climate and the earth's temperature by burning fossil fuels, cutting down forests and farming livestock that increases the concentration of atmospheric CO2 by more than 40%, with over half the increase occurring since 1970.. This adds enormous amounts of greenhouse gases to the atmosphere, increasing the greenhouse effect and global warming. At its most basic, climate change is caused by a change in the earth's energy balance how much of the energy from the sun that enters the earth (and its atmosphere) is released back into space. The earth is gaining energy as we reduce the

Greenhouse gases are released into the atmosphere at various scopes (Figure 6.1) that have an influence on the earth's energy balance by trapping heat in the atmosphere, which makes the Earth warmer. In Earth's atmosphere like carbon dioxide, methane, nitrous oxide and Hydrofluorocarbons (HFCs) act as a **greenhouse**, preventing a certain amount of heat radiation from escaping back to space.



Natural and anthropogenic substances and processes that alter the Earth's energy budget are drivers of climate change.

amount of solar energy that is reflected out to space.

What are greenhouse gases?

Greenhouse effect

Greenhouse gases (GHGs) are gases in the earth's atmosphere that trap heat. Water vapour, Carbon dioxide (CO2), methane and nitrous oxide are the major GHGs. During the day, the sun shines through the atmosphere and warms the earth's surface. At night the earth's surface cools, releasing heat back into the air. During this time some of the heat is trapped by the greenhouse gases in the atmosphere and keeps the earth's temperature at an average 14˚C (57˚F). The gases act like glass walls of a greenhouse and hence the name, greenhouse gases. Without this greenhouse effect, temperatures would drop to as low as -18˚C (-0.4˚F); too cold to sustain life on earth.

Figure 6.1. Source–Path relationship of Greenhouse gases

Since the Industrial Revolution, human activities have added very large quantities of greenhouse gases into Earth's atmosphere. These GHGs act like a blanket or car windshield to trap the sun's energy and heat, rather than letting it reflect back into space. When the concentration of GHGs is too high, too much heat is trapped, and the earth's temperature rises outside the range of natural variability. There are many GHGs, each with a different ability to trap heat and a different half-life in the atmosphere. GHGs are sometimes called "climate active pollutants" because most have additional notable effects on human health.

Table 6.1. Types of greenhouse gases and their major sources

GHG Categories	Major Sources
Carbodioxide (CO_2)	Fossil Fuel Combustion, deforestation
Methane (CH_4)	Landfills, Rice paddies, Digestive tracts of cattle and sheep
Nitrous Oxide (N_2O)	Fertilizer, animal waste
Hydrofluorocarbons (HFCs)	Semiconductor manufacturing and other industrial processes
Perfluorocarbons (PFCs)	Same as HFCs, plus aluminum smelting
Sulfur hexafluoride (SF_6)	Electrical transmission systems, magnesium and aluminum production

Carbon dioxide (CO_2) is the GHG responsible for greatest amount of warming to date. The majority of CO_2 is released from the incomplete combustion of fossil fuels-coal, oil, and gas used for electricity production, transportation and industrial processes (Table 6.1).

Carbon dioxide absorbs and radiates heat warmed by sunlight, and thermal infrared energy (heat). Unlike oxygen or nitrogen

Self-questioning

- What is the difference between global warming and climate change?
- Is the sun a major driver of recent changes in climate?
- What happens to carbon dioxide after it is emitted into the atmosphere?

which make up most of our atmosphere, GHGs absorb that heat and release it gradually over time and trap additional heat and raise Earth's average temperature.

Carbon dioxide is one of the most important elements of the earth's long-lived greenhouse gases. It absorbs less heat per molecule than methane or nitrous oxide, but it's more abundant and stays in the atmosphere much longer. Increases in atmospheric carbon dioxide are responsible for about two-thirds of the total energy imbalance that is causing the earth's temperature to rise (see Figure 6.2 and Figure 6.3).

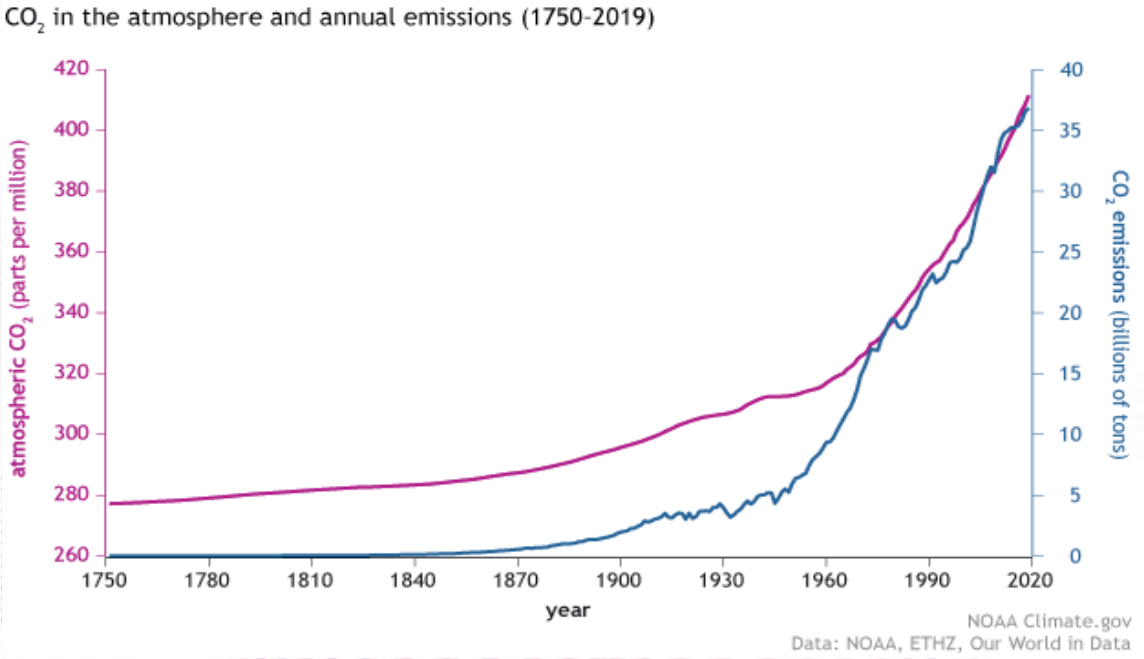


Figure 6.2. Progressive increase of of atmospheric Carbon dioxide concentrations since 1750 GC.



Figure 6.3. Ethiopia's annual CO_2 emission

In nature, CO_2 is exchanged continually between the atmosphere, plants, and animals through photosynthesis, respiration, and decomposition, and between the atmosphere and ocean through gas exchange. A very small amount of CO_2 roughly 1% emitted from fossil fuel combustion, is also from volcanic eruptions. This is balanced by an equivalent amount that is removed by chemical weathering of rocks. The CO_2 level in 2019 was more than 40% higher than it was in the 19th century. Most of this CO_2 increase has taken place since 1970 when the global energy consumption accelerated. The rise in CO_2 is largely from combustion of fossil fuels. Deforestation and other land use changes have also released carbon from the biosphere (living world) where it normally resides for decades to centuries. The additional CO_2 emitted from fossil fuel burning and deforestation has disturbed the balance of the carbon cycle because the natural processes that could restore the balance are too slow compared to the rates at which human activities are adding CO2 to the atmosphere. As a result, a substantial fraction of the CO2 emitted from human activities accumulates in the atmosphere, where some of it will remain for decades or centuries but for thousands of years. Comparison with the CO2 levels measured in air extracted from ice cores indicates that the current concentrations are substantially higher than they have been in at least 800,000 years.



Activity 6.3

Search from different sources (books, internet) about greenhouse gases and their source of emission, and effect. Present your finding to the class.

6.2. Effects of climate change

"You say you love your children above all else, and yet you are stealing their future in front of their very eyes." Greta Thunberg.



Figure 6.4 Greta Thunberg, Climate activist and Founder of Climate School Strike

Self-questioning

Can you explain some of the consequences of climate change?

Climate change destabilizes the earth's temperature equilibrium and has far-reaching effects on human beings and the environment (Figure 6.5). During the course of global warming the energy balance and thus the temperature of the earth change posing a significant impact on humans and the environment. And, it is likely that global warming will increase the probability of extreme weather events.

The direct consequences of man-made climate change include:

rising maximum temperatures

rising sea levels

higher ocean temperatures

an increase in heavy precipitation and shrinking of glaciers

thawing of permafrost

The indirect consequences of climate change which affect humans and our environment include:

- an increase in hunger and water crises, especially in developing countries
- health risks due to the rising air temperatures and heat waves
- economic crisis
- increasing spread of pests and pathogens
- loss of biodiversity due to limited adaptability of flora and fauna
- ocean acidification due to increased HCO_3 concentrations in the water as a consequence of increased CO_2 concentrations

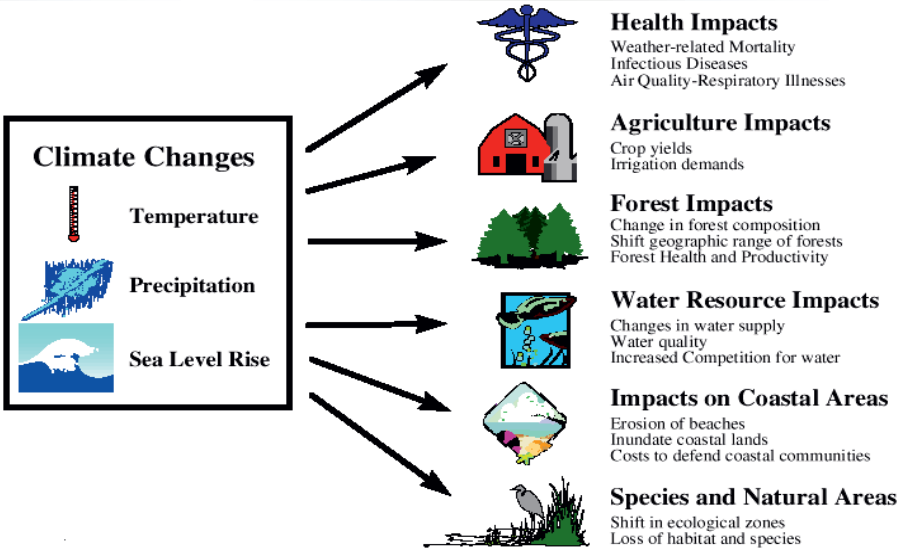


Figure 6.5. Potential climate change impacts

6.2.1. Effects of climate change on biodiversity

Biodiversity provides immense direct benefits to humanswith at least 40% of the world's economy being derived from biological resources. Maintaining biodiversity provides greater food security, opportunities for economic development, and provides a foundation for new pharmaceuticals and other medical advances. However, climate change is affecting the habitats (Figure 6.6) of several species, which they must either adapt or migrate to areas with more favorable conditions. Climate change is likely to become the dominantly direct driver of biodiversity loss by the end of the century. Projected changes in climate combined with land use change and the spread of exotic or alien species are likely to limit the capability of some species to migrate and therefore will accelerate their loss.



Figure 6.6 Changes in rainfall patterns can damage land, plants and animals

6.2.2. Effects of climate change on Agriculture

Agricultural biodiversity refers to all components of biological diversity of relevance to food and agriculture. It includes plants genetic resources, crops, wild plants harvested and managed for food, trees on farms, pastures, rangeland species, medicinal plants and ornamental plants of aesthetic value. Animal genetic resources include domesticated animals, wild animals hunted for food, wild and farmed fish and other aquatic organisms, insect pollinators and microbial and fungal genetic resources. Climate change can disrupt food availability, reduce access to food, and affect food quality. For example, projected increases in temperatures, changes in precipitation patterns, changes in extreme weather events, and reductions in water availability may all result in reduced agricultural productivity (Figure 6.5)

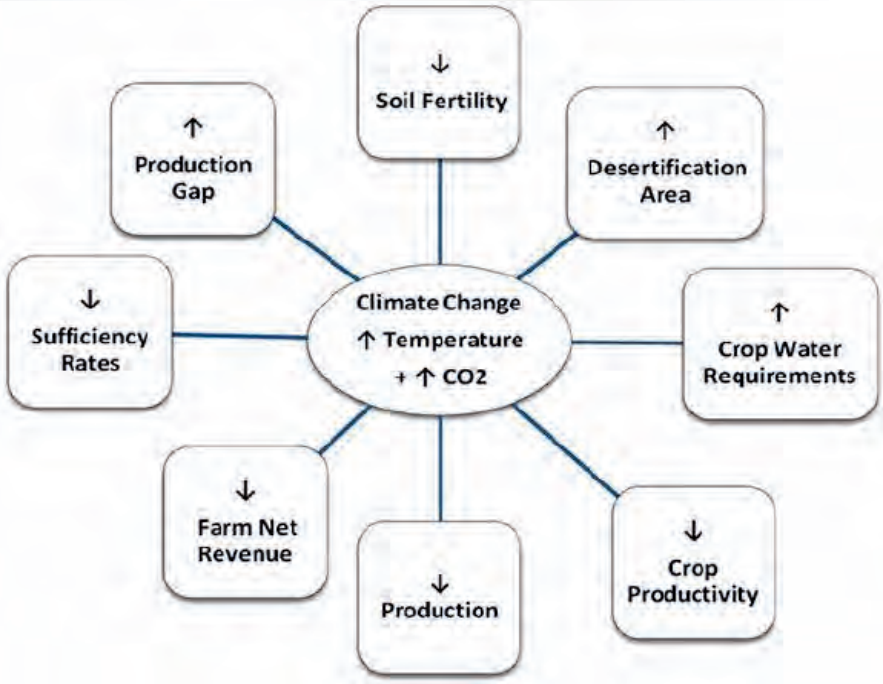


Figure 6.7 Climate change and its potential impacts on Agriculture.

6.2.3. Effect of climate change on forest productivity

Climate change could alter the frequency and intensity of forest disturbances such as pest outbreaks, invasive species, wildfires, and storms. These disturbances can reduce forest productivity and change the distribution of tree species. In some cases, forests can recover from a disturbance. In other cases, existing species may shift their range or die out. In these cases, the new species of vegetation that colonize the area create a new type of forest. Insect outbreaks often defoliate, weaken, and kill trees. Warm temperatures and drought conditions during the early summer triggered wild fire, which can consume millions of acres of forest (Figure 6.8).



Figure 6. 8 Galema forest on fire, Chilalo, Oromia state Ethiopia

6.2.4. Climate change and natural disasters

6.2.4.1. Climate change and natural disasters

Climate hazards are natural events in weather cycles. In our planet we always had hurricanes, droughts and wildfires, high winds and flooding (Figures 6.9 - 6.11). Surprisingly however, we are currently witnessing a scale of destruction and devastation that is new and terrifying. With the increasing global surface temperatures, the possibility of more droughts and increased intensity of storms will likely to occur. As more water vapor is evaporated into the atmosphere, it becomes fuel for more powerful storms to develop.



Activity 6.4

Discuss in a small group about the cause and effects of natural disasters such as flood, land slide and ice melt and the safety rules/ precautions to take during these natural disasters.



Figure 6.9. Heavy rains and storms caused damage and flooding to the Lietchuor camp in Ethiopia, Gambella region.



Figure 6.10 Dust storms in Eastern Ethiopia

6.2.4.2. Melting ice and rising seas

When water warms up it expands. At the same time global warming causes polar ice sheets and glaciers to melt. The combination of these changes is causing sea levels to rise resulting in flooding and erosion of coastal and low lying areas.

6.2.4.3. Extreme weather, shifting rainfall

Heavy rain and other extreme weather events are becoming more frequent. This can lead not only to floods and decreasing water quality but also to decreasing availability of water resources in some regions (Figure 6.11).



Figure 6.11 Failed crops as a result of climate change, Ethiopia

6.2.4.4. Risks of climate change for human health

Climate change is already having an impact on health: There has been an increase in the number of heat-related deaths in some regions of the earth and a decrease in cold-related deaths in other parts of the world. We are already observing changes in the spread of some water-borne illnesses and disease vectors.

Safety rules/ precautions during natural disaster

Disaster risk management is a comprehensive approach involving the identification of threats through implementation of the proposed mitigation measures (Figure 6.10). Natural disasters are catastrophic events that often occur without warning and disrupt the ecosystem causing damage to personal lives, property, transportation, and livelihood. While it is never possible to prevent a disaster, the damage can be minimized through timely preparation. As such, every natural disaster has its own set of precautions to be taken, which must be followed to save precious lives.



Figure 6.12. Elements of Comprehensive risk management

Here under are a few precautions to be taken during natural disasters.

i. Earthquakes

- The shifting of tectonic plates under the earth's crust causes earthquakes, which are responsible for mass destruction. When faced with an earthquake, these tips can be of use:

If you are indoors

- Take cover under a sturdy table or other pieces of furniture, and hold on until the shaking stops.
- Stay away from glass, windows, outside doors and walls, and anything that could fall, such as lighting fixtures or furniture.
- Stay inside until the shaking stops, and it is safe to go outside. Most injuries occur to people trying to move a different location inside the building or try to leave.
- Do not use the elevators.

If you are outdoors,

- Stay away from buildings, streetlights, and utility wires.

- Stand in open ground until the shaking stops. It's dangerous to stay directly outside buildings, at exits, and alongside exterior walls. Ground movement during an earthquake is seldom the direct cause of death or injury. Most earthquake-related casualties result from collapsing walls, flying glass, and falling objects.

ii. Tsunamis

- Tsunamis are a series of enormous ocean waves caused by earthquakes, underwater landslides, or volcanic eruptions. Tsunami waves range from tens to hundreds of feet tall and can travel twenty to thirty miles per hour. When faced with this phenomenon, these tips are to be followed:
    - Turn on your radio/TV to learn and follow the precautionary instructions during a tsunami warning, primarily when you reside near a coastal area.
    - Move inland to higher ground immediately and stay there.
    - Check for a noticeable recession in water away from the shoreline as this is nature's tsunami warning and should be heeded. It would help if you moved away immediately.
    - Stay away from flooded and damaged areas until officials say it is safe to return.
    - Keep yourself away from debris in the water; it may pose a safety hazard to boats and people.

iii. Cyclones

- Cyclones are tropical storms, caused by atmospheric disturbances around a low-pressure area. Cyclones are accompanied by strong winds, moving at a speed of sixty-two Kmph or more. When faced with a hurricane, keep these tips in mind:
- Be alert to the changing weather conditions.
- Listen to radio/TV for the latest information.
- Look for approaching storms.
- Look for the following warning signs: – Dark, often greenish sky – Large hailstones, a large, dark, low-lying cloud (particularly if rotating), roars, similar to a freight train. If you see approaching storms or any of the danger signs, be prepared to take shelter immediately. If you are under a tornado warning, seek shelter immediately. NOTE: In places where you have designated cyclone shelters, take refuge there.

iv. Floods

Floods are among the earth's most common and dangerous natural hazards formed due to a flow of water on areas of land that are usually dry. Excessive rains can damage nearby dams where tsunamis are some of its causes. When faced with flooding, these tips are to be followed:

- Do not attempt to walk, swim, or drive through the floods. Floodwater contains debris and contamination and can also be deadly due to fallen electrical lines in the water.
- Stay clear of bridges over fast-moving water.
- Keep an eye out for evacuation alerts.
- Move to higher ground. If your vehicle is trapped in flood and water starts filling inside the car, seek refuge on the roof.

Mitigation Measure

6.3. International conventions

Climate change is a long-term, global problem. Long-term problems generally require stable but flexible policy implementation over time. Various international conventions have continuously evolved to address the increasingly complex and changing environmental priorities of the world (Figure 6.15).



Activity 6.5

Make a small group and search international conventions from the internet or collect from nearby responsible offices and discuss whether these conventions are implemented as intended in the world and Ethiopia in particular. Report your finding to the class.

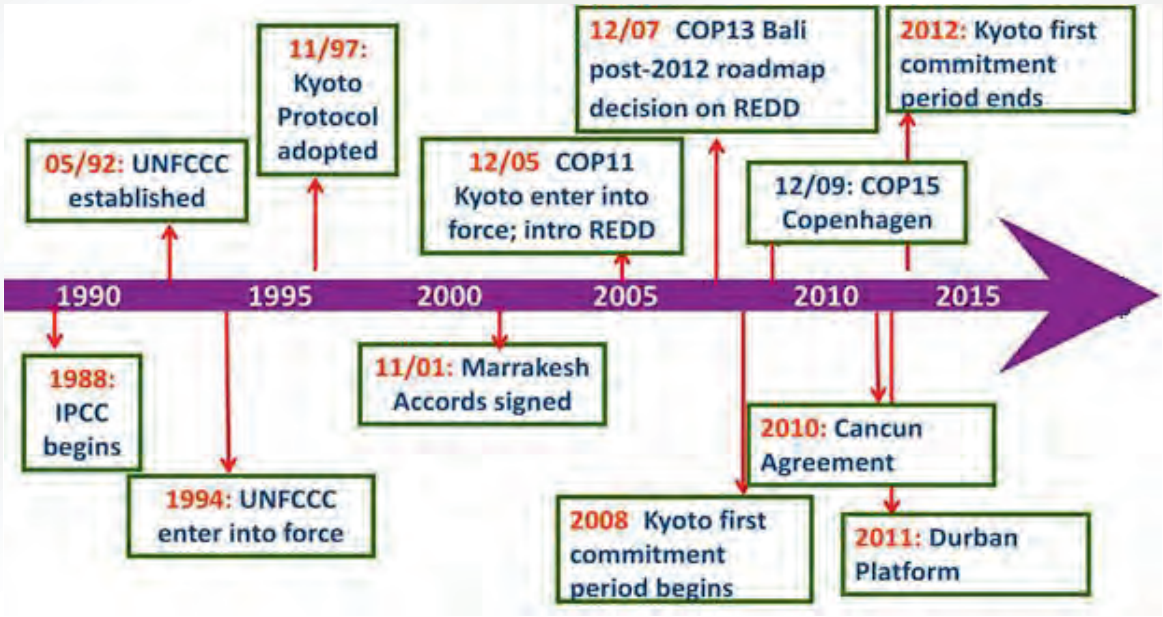


Figure 6.13. Climate Policy Timeline

In 2009 in Copenhagen (COP15), for the first time it was decided that each country would propose a national contribution (INDC, Intended Nationally Determined Contributions). All the 195 UNFCCC countries pledged to reduce their greenhouse gas emissions by 2025-2030.

It aimed "to strengthen the global response to the threat of climate change, in the context of sustainable development and efforts to eradicate poverty" and contained three key provisions:

- Holding the increase in the global average temperature to well below 2 °C above pre-industrial levels and to pursue efforts to limit the temperature increase to 1.5 °C above preindustrial levels, recognizing that this would significantly reduce the risks and impacts of climate change;
- Increasing the ability to adapt to the adverse impacts of climate change and foster climate resilience and low greenhouse gas emissions development, in a manner that does not threaten food production;
- Making finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development.

6.3.1. The United Nations Framework Convention

The United Nations Framework Convention on Climate Change (UNFCCC), agreed in 1992, is the main international treaty to combat "dangerous human interference with the climate system", in part by stabilizing greenhouse gas concentrations in the atmosphere. Its objective is to prevent dangerous man-made interference with the global climate system. The UNFCCC is an international environmental treaty. Ethiopia and all its member countries are among the 197 Parties of the Convention.

6.3.2. Kyoto Protocol on Climate Change

The Kyoto Protocol was adopted on 11 December 1997 The Kyoto Protocol operationalizes the UNFCCC by committing industrialized countries and economies in transition to limit and reduce GHG emissions in accordance with agreed individual targets. That means the Kyoto Protocol is an international agreement that called for industrialized nations to reduce their greenhouse gas emissions significantly. Other accords, like the Doha Amendment and the Paris Climate Agreement, have also tried to curb the global-warming crisis.

First agreed in 1997, it took eight years for participating countries to ratify the Kyoto Protocol (Figure 6.13). The deal was simple. Industrialized countries would be legally obliged to cut their greenhouse gas emissions 5% on 1990 levels by 2008-2012. Developing countries – including China, India, Brazil and South Africa – would face no restriction on their emissions but were encouraged to adopt policies to promote greener growth. To help countries meet targets, Kyoto also offered a range of market mechanisms that could help rich countries offset emissions by investing in low carbon projects in poorer parts of the world. It was hailed as an "environmentally strong and economically sound" deal by US President Bill Clinton, speaking just after agreement had been reached in 1997. "It reflects a commitment from our generation to act in the interests of future generations," he said.

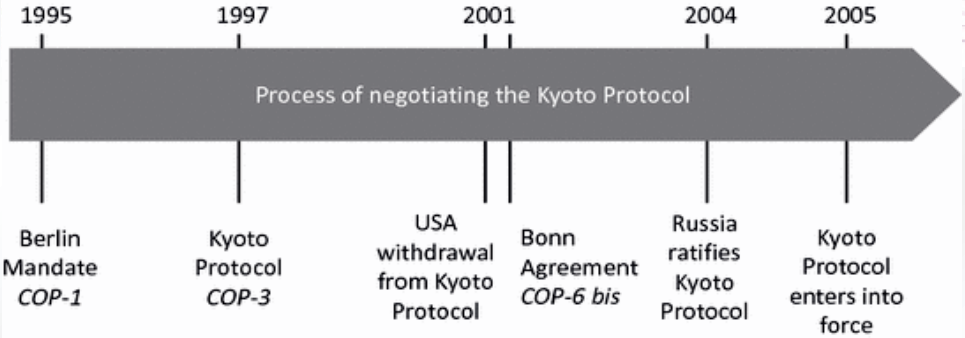


Figure 6.16. The Negotiation process leading to the Kyoto Protocol

6.3.3. International and national practices of Implementation of conventions

Ethiopia Green Legacy Initiave

Ethiopia is one of the world's most drought-prone countries. It has a high degree of vulnerability to hydro-meteorological hazards and natural disasters. Dependence on sectors that are climate change sensitive such as rain-fed agriculture, water, tourism, and forestry as well as a high level of poverty are the main factors that exacerbate Ethiopia's vulnerability.

Ethiopia's policy response to climate change has progressively evolved since the ratification of the UNFCCC in 1994. Ethiopia launched the National Adaptation Plan of Action in 2007 and the Ethiopian Program of Adaptation on Climate Change and Nationally Appropriate Mitigation Actions in 2010. Moreover, Ethiopia endorsed a Climate Resilient Green Economy (CRGE) strategy in 2011 to build a green and resilient economy. Over the years, Ethiopia has been implementing various programs within those policy frameworks. One among them, and by far the most consequential, has been the Green Legacy Initiative (GLI) launched in 2019 with a vision of building a green and climate-resilient Ethiopia and targeted to plant 20 billion seedlings within a period of four years. The Green Legacy Initiative is a demonstration of Ethiopia's long-term commitment to a complicated response to the impacts of climate change and environmental degradation that encompasses agroforestry, forest sector development, greening and renewal of urban areas, and integrated water and soil resources management.

Great Ethiopian Renaissance Dam (GERD)

The Grand Ethiopian Renaissance Dam (GERD) is a hydroelectric dam on the Abay (Blue Nile) River. The GERD is under construction since 2011 with the primary purpose of generating electricity with a capacity of 5.15 gigawatts to relieve Ethiopia's acute energy shortage and for electricity export to neighboring countries. The dam is located in the Benishangul-Gumuz Region and will be the largest hydroelectric power plant in Africa when completed. The GERD is one of éthe best example in Ethioia to support Climate Resilient Green Economy (CRGE) strategy to protect the country against the adverse effect of climate change and to build green economy.

Ethiopia exports electricity to Djibouti and to Sudan and has concluded power export deals and constructed electric transmission line

with Kenya and South Sudan. Ethiopia intends to generate foreign exchange through utilizing all of its potential for producing electricity, mostly from hydropower plants. The enormous amount of electricity generation hydropower like GERD will contribute to the region's economic integrity. Moreover, the lake being created on the GERD will be the biggest lake in Ethiopia with considerable potential for fishing and irrigation development.



Activity 6.6

1. Discuss in a group about the contribution of the Ethiopian Green Legacy Initiative campaign which was launched in 2019 in terms of international protocol implementations. Discuss further what is carbon sequestration, carbon trading and their link to the Green Legacy Initiative.

2. Discuss in a group about the contribution of GERD to support green economy and avert the adverse effect of climate change.

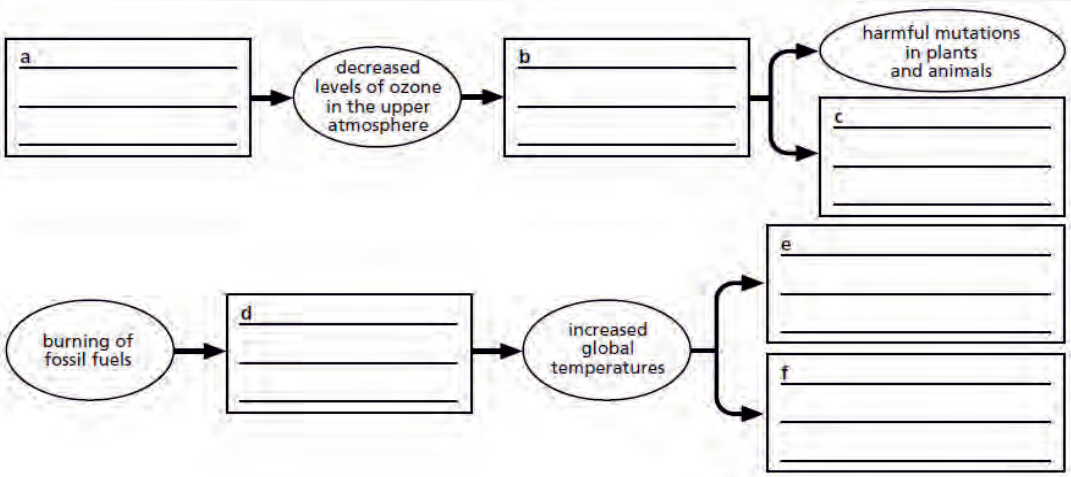
Unit Summary

In this unit you have learnt that climate change is the global phenomenon of climate transformation characterized by the changes in the usual climate of the planet which are largely caused by human activities. Natural and anthropogenic substances are the main causes of climate change. Human activities increasingly influencing the climate and the earth's temperature by burning fossil fuels, cutting down forests and farming livestock that increases the concentration of atmospheric Carbon dioxide (CO_2). Greenhouse gases (GHGs) such as water vapour, Carbon dioxide, methane and nitrous oxide are the major GHGs in the earth's atmosphere that trap heat. When Greenhouse gases are released into the atmosphere at various scopes they influence the earth's energy balance by trapping heat in the atmosphere, which makes the Earth warmer. Since the Industrial Revolution, human activities have added very large quantities of GHFs into Earth's atmosphere and has brought significant effect to the world. The effect of climate change has both direct and indirect consequences to humans. To tackle the effects of climate change several mitigation measures have been set globally. However, much work is expected from all countries to effectively implement international conventions and policies.

Unit review questions

Answer the following questions

1. What is climate change? Explain how it differs from global warming?
2. How does climate change affect human health?
3. How is agricultural productivity related to climate change?
4. Describe some of the climate change conventions
5. What is climate change mitigation?
6. The flowcharts below represent some of the effects of human activity on the environment. Each arrow indicates a known or suspected cause and-effect relationship. Complete the flowcharts by writing an appropriate response in the space corresponding to each box.


a
decreased levels of ozone in the upper atmosphere
b
harmful mutations in plants and animals
c
burning of fossil fuels
d
increased global temperatures
e
f